

Nicola Kunz

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Education

Northeastern University – GPA 3.608/4.0 Expected August 2027

Master of Science in Mechanical Engineering with Focus in Mechatronics

- Relevant Courses: Robot Mechanics and Controls, Control System Engineering, Regulatory and Quality Aspects of Medical Device Design, Mechatronic Systems

Rochester Institute of Technology – GPA 3.19/4.0 December 2023

Bachelor of Science in Electrical Mechanical Engineering Technology

- Relevant Courses: Quality Engineering Principles, Automation Control Systems, Control Systems, Mechanics for Mechatronics, Manufacturing Processes, Introduction to Digital & Micro Systems

Skills

Software: MATLAB, Simulink, SolidWorks, Minitab, JMP, C++, C (Embedded/Zephyr)

Engineering: DFM/DFA, GD&T, Mechatronics, Precision Mechanisms, Testing & Validation, Technical Documentation

Manufacturing: 3D Printing, CNC Machining, Waterjet Cutting, Precision Fabrication, Embedded Systems (nRF7002)

Certifications: Six Sigma Green Belt

Languages: English (Fluent), German (Native)

Soft Skills: Problem Solving, Technical Communication, Teamwork, Adaptability, Attention to Detail

Work Experience

EXP Makerspace Student Supervisor, Northeastern University – Boston, MA January 2026 – Present

- Oversee 8 woodworking machines as sole supervisor ensuring safe operation for 3-4 daily beginner patrons
- Execute daily calibration for CNC and waterjet; troubleshoot 3D printers to minimize downtime; advancing toward equipment trainer certification

Engineering Intern, Volpi USA – Auburn, NY September 2024 – January 2025

- Engineered precision fixtures ($\pm 0.05\text{mm}$) for clinical medical devices applying DFM/DFA; produced GD&T SolidWorks drawings
- Implemented 3D printing reducing costs **\$400/unit (67%)** and lead time by 2-3 days; reverse engineered 15 fixtures generating CAD models accepted for manufacturing
- Validated 3 prototypes for dimensional accuracy, refining designs to meet assembly requirements

Engineering Intern, Volpi CH – Dietikon, Switzerland April 2022 – July 2022

- Developed GD&T-compliant CAD drawings for medical components; machined precision prototypes using CNC, drill press, and lathe

Engineering Intern, Volpi USA – Auburn, NY April 2021 – July 2021

- Reverse engineered clinical equipment identifying suppliers and **lowering costs by 20%**; generated wiring diagrams and established CAD guidelines adopted across projects

Technical Projects

LiverLugger™ Organ Transport System – Livera Dynamics (Capstone) Spring 2026

- Served as Manufacturing Leader and Regulatory Affairs lead for a Class IIa organ transport device targeting FDA 510(k) clearance and CE Mark under EU MDR 2017/745
- Built Design History File spanning **66 controlled documents**: GSPR checklist (EU MDR Annex I), thermal verification test protocol, and DHF index; coordinated design inputs covering IEC 60601-1, IEC 62304, and IEC 62366

Patient Monitoring System – 1st Place (7 Teams), Nordic Semiconductor IoT Hackathon January 2026

- Built real-time patient monitoring system with 2-person team integrating ECG, temperature, and humidity sensors with LED triage alerts and centralized web dashboard on unfamiliar nRF7002 hardware in 48 hours
- Implemented Bluetooth peripheral/central architecture for multi-patient data aggregation and Wi-Fi cloud transmission; **0.5-second** sensor refresh rate

Precision Mechatronic Gripper Fall 2025

- Engineered dual-mode gripper with 6-bar linkage achieving **sub-1mm manipulation** (40-45g capacity) at $\pm 0.01''$ tolerance; refined reliability across **3 prototype iterations** through custom fasteners and material optimization